NAZMUL HUDA BADHON

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EDUCATION

Bachelor of Science in Computer Science and Engineering *Jan 2022 – Jan 2026*
Daffodil International University

- CGPA: **3.88** out of 4.00 (**Top 5% of the cohort**)
- Received **90% tuition fee waiver** for merit-based scholarship
- **Notable Courses:** Data Structure, Algorithm, Programming and Problem Solving, Discrete Mathematics, Data Communication, Object Oriented Programming, Database Management System, Compiler Design, Numerical Methods, System Analysis & Design, Operating Systems, Software Engineering, Computer Graphics

Exchange Semester in Computer Engineering *Sep 2023 – Dec 2023*
Dongseo University, South Korea

- GPA: **3.94** out of 4.50
- Received **GKS scholarship (**fully funded by the Government of the Republic of Korea**)** for Fall 2023
- **Notable Courses:** Artificial Intelligence, Mobile Programming, Cloud Computing, Computer Networks

WORK EXPERIENCE

Graduate Researcher (Part-Time) *Jan 2026 - Present*
CSE AI Centre, Daffodil International University

- Worked under the supervision of Dr. Fernaz Narin Nur, PhD, and Nazmun Nessa Moon gaining valuable mentorship in advanced research methodologies.
- Contributed to the drafting and peer review of academic journals focused on image processing and classification, ensuring clarity, rigor, and technical accuracy.

THESIS

Tabular data-based lightweight convolutional neural network for electricity energy demand prediction

- Developed **TLCNN**, a lightweight convolutional neural network designed to process structured tabular data directly for electricity-demand forecasting, eliminating the need for complex tabular-to-image conversion.
- Built a robust preprocessing pipeline for the **BanE-16 energy dataset**, incorporating missing-value treatment, feature selection, standardization, target normalization, Gaussian-noise augmentation and CNN-compatible data restructuring.
- Evaluated against traditional machine-learning baselines, including Linear Regression, Random Forest, and XGBoost, achieving a 10.89% lower mean squared error and demonstrating improved predictive accuracy and generalization.

PUBLICATIONS

2026 Temporal Convolutional Network-Based Indoor Location Recognition from BLE Beacon Signals in a Nursing Care Facility. ***1st Author*** *(IJABC, Kyushu Technology University, Japan)* View Full Paper View Full Paper

2026 Bridging Neuroscience and Education: A Scoping Review of EEG-Based Cognitive and Emotional States in Educational Settings. ***1st Author*** *(IJEDRO, Elsevier, Scopus Q1, Cite Score:6.1)* **Under Review**

2026 Predicting Peak Energy Demand Using Machine Learning Techniques for Efficient Electricity Supply Management. ***2nd Author*** *(Book Chapter in Taylor & Francis)* View Full Paper

2025 TLCNN: Tabular data-based lightweight convolutional neural network for electricity energy demand prediction. ***1st Author*** *(Global Energy Interconnection, Elsevier, Scopus Q1, IF:2.6)* View Full Paper

2025 A comparative deep learning methodology for plant insect image classification: assessment of CNN architectures and augmentation techniques. ***2nd Author*** *(MethodsX, Elsevier, Scopus Q1, IF:1.9)* View Full Paper

2025 Systematic Literature Review of LLM-Large Language Model in Medical: Digital Health, Technology and Applications. ***Co-Author*** *(Engineering Reports, Wiley, Scopus Q2, IF:2.0)* View Full Paper

2024 BAU-Insectv2: An Agricultural Plant Insect Dataset for Deep Learning and Biomedical Image Analysis. ***Co-Author*** *(Data in Brief, Scopus Q1, IF:1.4)* View Full Paper

PROJECTS

Breast Cancer Segmentation and Classification Using Hybrid Deep Learning

- Developed a two-stage breast ultrasound analysis pipeline using U-Net to segment regions of interest, followed by classification of benign, malignant, and normal cases from the segmented images.
- Constructed a hybrid VGG16–ResNet50–MLP model through feature concatenation, achieving 95.94% test accuracy, 95.93% F1-score, and 0.94 MCC across 123 test images.
- **Project link:** *https://github.com/nazmulhudabadhon/Computer-Vision-and-Deep-Learning-in-Medical-Imaging/*

Lung Histopathology Classification Using Image Enhancement and Transfer Learning

- Built an image-enhancement and transfer-learning pipeline using threshold-based segmentation and pretrained models including VGG16, ResNet50, EfficientNetB0, InceptionV3, and MobileNetV2.
- Demonstrated the strongest performance with VGG16, reaching 80% test accuracy and 0.80 macro F1-score on three lung tissue classes.
- **Project Link**: *https://github.com/nazmulhudabadhon/Computer-Vision-and-Deep-Learning-in-Medical-Imaging/*

Temporal Convolutional Network–Based Indoor Location Recognition Using BLE Signals

- Built a TCN-based indoor location recognition system using BLE RSSI data from a real nursing care facility. Developed a robust preprocessing pipeline for noisy and imbalanced sensor data.
- Implemented causal dilated convolutions with residual connections to model temporal dependencies across 22 indoor locations. Achieved 66% accuracy and 0.56 Macro F1-score, outperforming non-temporal baselines and improving minority-class recognition.
- **Paper Link:** *jstage.jst.go.jp/article/ijabc/2026/2/2026_154/_article*
- **Project Link:** *https://github.com/nazmulhudabadhon/deeptrack-ble-indoor-location-tcngnition-Using-BLE-Signals*

SKILLS

Programming Skills: C, Python, Java
Machine Learning Skills: Pytorch, TensorFlow, Scikit-learn
Data Analysis: NumPy, Matplotlib, Seaborn
Software Skills: Microsoft 365, Mendeley Reference Manager, LaTeX, Adobe Photoshop, Adobe Illustrator, Figma
IELTS: Band 6.5 (L-6.5, R-7, W-7, S-6.0)

EXTRA-CURRICULAR ACTIVITIES

2024 **Participant,** Crack Dataset Contest, Daffodil International University View Event
2024 **Volunteer Intern,** International Affairs, Daffodil International University
2024 **Participant and Cultural Performer,** K-Culture Lover's Night, Embassy of the Republic of Korea View Event
- Represented DIU through a Bangladeshi musical performance during a cross-cultural program.

2024 **Founding General Secretary,** Daffodil International University Research Society
- Coordinated research activities, events, and member engagement.
- Helped promote a collaborative research culture among students.

2023 **Team Leader,** International Camp, Dongseo University, South Korea
2023 **Event Volunteer,** Multi Destination Education Expo, Admission.ac View Event
2022 **Take-Off Programming Contest,** Daffodil International University View Event
- Solved **3** out of 8 problems, achieving **111th** Rank out of 350 contestants

CERTIFICATIONS

2025 **Research Award,** Division of Research, Daffodil International University View Certificate
- In recognition of **five** publications in reputed indexed journals

2025 **Peer Reviewer,** Digital Health (Journal), Sage Publishing View Certificate
2025 **Top Performer,** Computer Vision and Deep Learning for Medical Data Analysis Bootcamp, HIRL, DIU View Certificate
2024 **Research Award,** Division of Research, Daffodil International University View Certificate
- In recognition of **one** publication in reputed indexed journal